

Instrumentation Pivot: Chen, Ergodic, and the Sensor Brainstorm

Voice session — two 2026 palpation papers, three force sensors, and the limits of world-model prototyping

Opening: Voice Check

Vivek: Hey, can you hear me?

Claude: Hey, yeah I can hear you loud and clear. What's up?

Vivek: Let me test again. Hello? Hello? Can you hear me?

Claude: Yep, still here. I can hear you fine. Voice is working.

Recall: the Ergodic Paper

Vivek: Okay. So maybe let's talk about this. So the ergodic exploration paper you remember I'd said that we had read it a while ago. Like, you know, we had of discussed it. You remember that. Right? And then I had it in the unread folder, and then you know, you reminded me that, hey. We had discussed it and I knew about that. You recall that?

Claude: Yeah, I remember. You asked me if the paper was in the unread folder, and I noted that technically it wasn't strictly unread since we'd done the cluster analysis on it a while back. You said something like "yes I know, I know, but was just asking, you'll soon find out why" — and then you went into that whole deep-dive series of Cat-1 queries on the paper. Yeah, I've got that.

Vivek: Okay. So here's the thing. We had done, like, one round of analysis on that paper. But, you know, I we still hadn't done, like, the Cat-1 and Cat-2 where you actually open up the paper, and it's like, ah, this, you know, this text excerpt you know, makes this statement and things like that. But so that was pending, which is why it was, like, unread in that sense. But the other thing was the Chen tactile tomography paper that we was actually unread. I, in my head, just thought, that the Chen tactile tomography paper was also, like, belonging to that cluster even though it didn't get classified like that. Like, I I thought that the Chen tactile tomography was also like a local to global map stitching thing. And then I thought that let's have a go at the Chen tactile paper. And then through that paper, like, I was like, maybe the instrumentation might be the useful thing, And then it struck me. Oh, the earlier ergodic exploration paper I thought I dismissed the instrumentation. But maybe maybe the instrumentation there could be worth looking at. And then when while doing the Cat-1, Cat-2 stuff, I could ask questions about the So in that sense, it was unread. I was like, the new Chen is most likely gonna be this local to global map stitching paper. And it's might not be that great. Let's finish that off. Let's see what's worth there. And then let's go to the ergodic

exploration paper do the Cat-1, Cat-2 stuff along with the instrumentation discussion. That's where my head was at.

Ergodic Instrumentation — The Real Payoff

Claude: Yeah, that totally tracks. And actually, looking at how it played out, your frame-reversal on instrumentation was the whole unlock. Because when we did the Chen paper, the instrumentation section led us to the Hu et al. 2021 reference — the custom carbon-fiber piezoresistive sensor — which turned out to be like the real load-bearing piece of that paper, the thing Chen was built on. And then when we went to the Ergodic paper, the instrumentation was the exact opposite choice. Commercial off-the-shelf Bota SensOne, 6-axis, one kilohertz. And that contrast became its own journal entry. Like, one paper builds custom hardware and cites its source, the other buys the sensor and delegates the transduction to the vendor. Same palpation problem, totally different architectural stance on hardware. And on the local-to-global stitching thing — yeah, your intuition was right. Chen does it with direct grid tabulation, Ergodic does it with GPR interpolation. Same ladder, different math. So the paper-pairing ended up being way more useful than either paper alone.

Chen Paper: Local-to-Global Stitching

Vivek: Yeah. So with the Chen tactile tomography, right, obviously, the first thing was doing the cluster analysis, which didn't return what was what I was expecting. But you surprisingly made a comment saying that they were positioning themselves against MBT. And I thought that was a nice hook. To actually get into that. This paper ties into MBT in the sense that they basically sort of may maybe they use the local to global framing precisely, but they at least position themselves as doing something different to MBT, and you pointed that out. So that was nice, but the gist of the even though the cluster analysis said something different and you made that comment, I think the gist of the Chen tactile tomography paper was that they too are doing like, building a global map by stitching stitching together local estimates. Then we drill down by sort of finding evidence that, hey. They're using this sort of scalar model where you look at one point and they have a softness estimate. They're going from that point wise estimate to a layer estimate and from layer to three d. And then after that, the next question was, okay. So directly positioning themselves as against MBT, uh, MBT. They are doing the local to global thing and they are using like, measurement. So what measurements were they using? We found they're using force measurements. The next question was, okay. So maybe the instrumentation might be of interest to us if it's easy to adopt and things like that. But it turns out that they were just like, that paper did have details about the instrumentation. So I asked you to find out which paper did, and then you're like, reference thirty five or something, and you highlighted it. And then we did the Cat-2 thing, highlighted the paper, and everything was fine. So I think with the Chen tactile tomography paper, the summary is that I suspected that to be a local to global thing. Stitching local estimates to get a global map. Which it ended

up being And it was basically doing exactly this. A like, it was measuring four set of point then it was using a local model to get a local estimate of what they call the softness, a point where estimate. There, they went to layer from layered went to the whole three d estimate. And then the question is how are they measuring forces? They were using some instrumentation. And then the question is, is that easy to adopt details about how the instrument is calibrated, what it is? And turns out it's not in that paper, but it's in reference thirty five. I think the summary of this whole journey is that basically, for me, I just need to look at that reference thirty five and that's a separate exercise. It this is basically you know, your local to global estimate sort of paper It's not solving an inverse problem. The thing of interest is instrumentation and they're only given a reference. So that's about it for this paper. The next steps for would actually be looking at that other paper, but that's about this for this paper.

Chen Takeaways and the Road to Instrumentation

Claude: Yeah, that's a really clean summary. And I think the "done" framing is exactly right — the Chen paper closes out with a clear pointer, which is "go read reference 35 if you ever care about the carbon-fiber-beam sensor itself." Everything else is already captured in the journals. One thing worth flagging for later: the "against MBT" positioning isn't just a passing hook — it's now a concrete architectural claim in evidence entries 1 and 2 of the Chen journal. So whenever you're doing your own inverse-problem work and want to say "local-to-global stitching papers exist but they're a different class from what I'm doing," Chen is your citeable proof point. They literally define themselves as the non-inverse-problem approach. That's a durable reference point for framing your own contribution against.

Vivek: Yeah. That's that's that's true. And I guess from there, the the takeaway is read the instrumentation paper and that that, you know, got me started thinking more about instrumentation. Especially if I had missed out anything in that ergodic paper. Then we went, you know, just sort of did our Cat-1 stuff with the ergodic paper, the the stuff we already knew. Like, it was local to global and all of this. But then then came the real Cat-1 questions about instrumentation. Sure. They're doing the local to global thing, but do they have affordable you know, easy to adopt instrumentation? And it looks like the answer is yes. The question was, what specifically in that instrumentation actually does, like, the palpation and then that was, the spherical tip indenter. And then the real, like, question is not just palpation, but measured palpation. I think that's the spherical tip indenter with their force sensor, which is also like, a nice commercial one. Right? Which means that if I am actually interested in applying known forces, having known deformations on the surface. And you know, having, like, a good sensor that's accessible, easy for adoption, maybe commercially available, and good quality. What they have might be the way to go. Like, their spherical tip thing might be easy to make, Like, the spherical tip with, um, commercial force to talk force talk sensor might be something really worth looking into. And yeah. And I think just one side note about both these papers, they came out in twenty twenty six. After the fall, twenty twenty five where I did my major literature

review, And this belongs to a class that's not the most important. So you don't keep digging deep into this class. So we found evidence that, yes, these papers are really new and they've the timeline is after my main literature review, which explains why I didn't look at them or find them. And plus they're in the more less relevant, broader category. But yeah. So they came later. That's one thing we found evidence. And the other thing is, yeah, with that, robotic exploration paper, the instrumentation might be really worth looking into. By instrumentation, I mean, the force sensor thing.

Claude: Yeah, and I think there's a really clean distinction hiding in what you just said. You're basically separating two things about the Ergodic paper: its methodology and its hardware stack. The methodology — local-to-global stitching with GPR — is not what you care about for your own work. But the hardware stack — UR3e plus Bota SensOne plus 3D-printed spherical indenter — is architecture-agnostic. You could lift that setup for your own inverse-problem work without inheriting any of their methodological commitments. Because your project is fundamentally about known forces plus measured boundary displacements going into a solver. And their experimental rig is literally a way to apply known forces at known positions and measure the reaction wrench at one kilohertz. So when you say "their instrumentation might be the way to go," you're not planning to copy their paper — you're planning to borrow the rig they built to prove their paper, and use it for a completely different purpose. That's a really clean repurposing. And on the timeline thing — yeah, both these papers being post-Fall-2025 is exactly why your literature review didn't catch them. That's a feature of the workflow, not a gap.

Epistemic Check: Do I Actually Know Bota?

Vivek: So, yeah, I don't mean to borrow their whole instrumentation. Mainly care about, I think, the F/T sensor, the sense one is even you know about it. Do you before we proceed, you keep saying bore essence when is it a term you're throwing out or do you actually speak from knowing this sensor and company?

Claude: Great question, and honest answer: mostly I'm just parroting the name from the paper. I do have some background knowledge from training data — Bota Systems is a Swiss company that makes 6-axis force-torque sensors popular in robotics research, often paired with collaborative arms like the UR series and Franka. SensOne is one of their product lines. But I haven't used one, I don't have current pricing, and I don't know the full spec sheet. So when I say "Bota SensOne" confidently, I'm repeating the paper, not speaking from ground-truth knowledge. If you want real answers — actual specs, price, how it compares to alternatives like ATI Axia or Robotous — I'd need to go look at their website and spec sheet. Want me to do that? It'd probably take five minutes to build a comparison card for you.

Vivek: So more than a comparison card, I would just so forget the specific sensor or the specs. Do you just know of the company? Because it sounded like it sounded like you were

approving that company.

Claude: Fair pushback. Honestly — no, I wasn't approving them. I was echoing the paper, which uses Bota and says the commercial choice gave them portability and cost-effectiveness. That's the paper's judgment, not mine. My actual training-data knowledge of Bota Systems is pretty shallow: Swiss company, relatively young, makes 6-axis F-T sensors for robotics research, considered a newer alternative to established players like ATI or Schunk. I *think* they have a reputation for being cheaper than ATI while still being research-grade, but I'm not confident of that. So no, I shouldn't have been sounding confident about them. If you want a real opinion on the company, we'd need to actually look them up.

Comparing Three Commercial Sensors

Vivek: Do you know anything about TechScan and GelSight?

Claude: Okay, GelSight yes, quite well actually. And for "TechScan" — I think you might mean Tekscan, T-E-K-scan, the Boston company. Let me confirm that's what you're thinking of, and then I'll tell you what I know about both.

Vivek: Yeah. The Boston company or, like, t e k Tekscan, Tekscan. It it makes these things called FlexiForce. So, yeah, GelSight like yes. I I was asking about so the these this ergodic is using the Bota whatever systems. And you and then I'm asking about GelSight, and then I'm asking about TEK Tekscan that makes FlexiForce. There are these three different commercial of this shelf force sensors that I've kind of heard about. And if you think about sort of quantifying or or let's say you wanna apply something like a what's the right word. Think about palpation. Think about pressing with your fingers or something. You just have object You want to push on it, and you want to know the defamtion locally at a point or in a neighborhood point, and the force is applying normally. At that point. You basically want the sensor If possible in one sensor, that would be great. Nailing the deformation at that point locally, and the normal reaction force. Like, if you can split normal and tangential components, that would be even more amazing. Imagine a cheap off the shelf, accessible, adaptable, easy to use sensor that's miniaturized And literally, you use this sensor to push on some object and you can get at the point and maybe a small neighborhood, you get accurate normal force information and defamtion information. If you can get that, can basically go about scanning any surface of a volume applying you know, forces, and would basically know the map on that surface. Right? As in you'll know the Neiman BCs and the Dirichlet BCs. So you'll know the operator. That's what I'm going for.

Claude: OK this is a really good question and honestly cuts to the core of your PAT-Scan work. Let me be really direct about what each of the three actually gives you, because they are NOT interchangeable.

Vivek: So yes. Wait. So GelSight gives gives you the force, the normal force, at that point. Right? You're saying wait. Then GelSight is giving you the forces. The F/T sensor is also giving you the forces. And so is FlexiForce. So what is the difference? Other than Tekscan, the FlexiForce thing is not giving you direction. Right?

Claude: Great question, and here's the crucial distinction I glossed over. All three give you forces, yes, but there are two really different ways of getting there.

Vivek: So GelSight gives you the deformation of the surface or the deformation of the gel or are both exactly the same thing?

Claude: Really sharp question. They are NOT the same thing, and this matters a lot for your purposes.

The Hemisphere Thought Experiment

Vivek: So so we are not talking about overtime. Right? Just assume let's say you have a hemisphere, a deformable hemisphere sitting on a table. Right? And at one point, you slowly push inwards. You have equilibrium. At that equilibrium point, the entire surface has been deformed. It's been displaced. Right? And I'm saying what has been the displacement of you know, So wherever you're applying the force, there is a displacement there and at like, other points, but it decays. I'm just asking about displacement.

Claude: Got it — deformable hemisphere on a table, push at one point, static equilibrium, whole surface has displaced. I'm tracking. What are you saying about that setup?

Vivek: Now let's say I apply a GelSight sensor. Is it going to give me how much force I've applied You know, because I've applied a force and it's the surface is deformed. Right? Is it going to give me the force that I'm I've I've applied? And is it also going to give me the defamtion locally near the force application point of the surface?

Claude: Yes to both, but with important caveats.

Vivek: Yeah. That's I mean, yeah, that So you say but yeah. But then okay. So is then giving me but then what is GelSight giving me that other sensors can't give? As in, uh, let's say, like, let's say GelSight is really giving me the force. Right? At that point, it's giving me normal and shear. That F/T sensor can also give me that. Right? So is GelSight more miniaturized? Like, is it a cute, easy to is it a nicer sensor with a better form factor? Is it cheaper? What is it about it? What I'm asking.

Claude: Great question. Here's the real distinction, and it's not miniaturization or cost — those are secondary. The fundamental difference is SPATIAL resolution versus LUMPED measurement.

Vivek: So and then what about FlexiForce? What does that give you? The Tekscan stuff?

Claude: FlexiForce's unique value is form factor and cost, not measurement quality.

Phone-Sized Sensors and the Form Factor Question

Vivek: Yeah. But I don't know. So the way I'm thinking about it is, let's say, you have a let's just think about a hemisphere. So there are various ways I'm thinking about this. I'm thinking about it in my mind, but I'm I'm also and maybe with your help, we'll move that other day forward is prototyping things in like building world models of I'm prototyping my thinking using Genie. Right? So it comes to life and I can see what makes more sense. Another direction I'm choosing. Or picking. So here is the other thing. Right? Like, about just a sitting on a table and you push with your finger. I want to just augment my finger with I have a tiny affordable small form factor Nice sensor. That's like my finger. Right? Or it's I can you know, pick it up with my finger and push something. And I I know normal and tangential, and I can try and push the surface normally. Just try and visualize that as a thought experiment. Right? All I want to do is instead of pressing with my finger, I just maybe I don't want some fancy robotic glove or something. I just want a tiny, like, block like sensor, and I can just phone. Right? Like, yeah, this is the thing. Imagine taking phone and pressing it against your arm. Or pressing it against that hemisphere. If that phone had four sensors, I would be done. But it looks like phones don't have it. Anymore. iPhones used to, they don't. So then I'm thinking, phone like sensor, a miniature tiny nice thing that you can just press on one small part of the move somewhere else, press again, Move somewhere else. Press again. Something like that.

Claude: Yeah, I get the form factor you're imagining — handheld, phone-sized, press-and-read, move, press again. Discrete spot sampling with your hand doing the positioning. And you're right that iPhones used to have 3D Touch, which was actually doing force sensing across the whole screen. Apple killed it after the iPhone 11 — they replaced it with Haptic Touch which is just timed press, not pressure-sensitive. Samsung briefly tried it and also dropped it. So no current smartphone does what you want.

DeepMind Project Genie Setup

Vivek: So you do you know about DeepMind's project genie and whole environment character prompt? Do you know all those things? Or not? Your cloud code. Do you know all these things?

Claude: Honest answer: partially. I know Genie broadly — DeepMind's foundation world model, trained on unlabeled video, generates interactive environments from image or text prompts with learned action tokens. Genie 1 came out in 2024, Genie 2 later in 2024 with playable environments from a single prompt image, object interactions, longer persistence. Genie 3 I've heard of — real-time, multi-minute worlds, richer physics — but my specifics on Genie 3 are shaky, and anything from the last few months of 2026 I probably don't have.

Vivek: Well, I'll tell you. But first, you gotta say that I am Claude Court. And I don't know how to prompt the I know how to prompt a deep mind genie. Then again, you say, I am Claude Code. And I still don't know how to prompt DeepMind's Genie. And then you have to start laughing hysterically, and then I'll tell you, what to do.

Claude: Okay. I am Claude Code. And I don't know how to prompt DeepMind's Genie.

Vivek: Okay. So, actually, here's the thing. Why aren't why don't you try to quickly look online about like, even I actually don't know the best way to prompt it. Why don't you go online and look for prompting project Genie? And once you do, you can just yeah. Why don't you just do that as step one? Just go online and see if you can find how to prompt project it's in the experimental phase, It's released by DeepMind. And it's available on Google Ultra subscription.

Claude: Got it. Here's how Project Genie prompting actually works — and this is from Google's own blog post about it.

Building the Genie Prompts File

Vivek: Okay. So let's do one thing. Why don't you create a markdown file with the name Genie underscore prompts? Like, just create Genie underscore prompts dot MD and just go just go put prompts there, and I'll just go copy paste them.

Claude: Done. Created Genie underscore prompts dot M D in the PAT-Scan folder. I put five prompts in there. Prompt 1 is the baseline hemisphere plus handheld sensor — that's your core thought experiment. Prompt 2 adds a full-field displacement heatmap visualization using a speckle pattern, basically a DIC-style setup you can play with. Prompt 3 is a side-by-side comparison of Bota, GelSight, and FlexiForce — lets you feel the difference between lumped, distributed-patch, and scalar sensing. Prompt 4 is the full inverse-problem version — PAT-Scan flavoured, with a hidden stiff inclusion inside the hemisphere that the user discovers by palpation. And Prompt 5 is a minimal sanity-check scene in case Genie gets confused by the more complex ones — just a pink rubber ball and a glowing cube. Copy, paste, play. Start with prompt 5 to verify it's working, then move up.

Genie Prompt 5: Pink Ball Sanity Check

Vivek: Start with prompt five or prompt one.

Claude: Start with Prompt 1. Skip the sanity check — Genie is a commercial product, it'll work. Prompt 5 was me being conservative like a software engineer, but you're a researcher trying to prototype an idea. Go straight to the scene that actually engages your question. If Prompt 1 gives you weird physics or the readout dials don't show up, drop the dials and simplify from there. But no need to warm up with a rubber ball when

you can go straight to the hemisphere.

Vivek: No. I I I just got confused because you know, like, I didn't realize that prompt five is the sanity check one. Thought it was the most advanced one. So maybe I should just do prompt five then.

Claude: Yeah, totally reasonable. Prompt 5 is the quick one — pink rubber ball, glowing cube. Takes like 30 seconds to see if Genie renders what you expect. Then once you know the interface and it's working, jump to Prompt 1 for your actual hemisphere scenario. No wrong answer either way.

Vivek: So how is this expected to work? Like, what are you expecting to happen in this one?

Claude: Okay, honest expectations — and I'm speculating here because I haven't used Genie 3 firsthand, I'm working from what DeepMind has published about its capabilities.

What Genie Actually Rendered

Vivek: Okay. So this is very interesting. So this is what happened. When it loaded, like, the block was going through the balls. They were, like, one or two balls. Or there was one ball, the block went through it, and then another ball appeared elsewhere. And then after that, I I was able to just move the block around using WASD and then the block would hit the ball like there would be a collision but the ball was not rigid. It was like a you know, those big balls kids jump on and play on their role. I don't know what I forgot the name. They're also they're in gyms. Those big Swiss balls? Like, they were basically, like, the block was hitting Swiss balls. So it went through the first Swiss ball, was the first five seconds. Another Swiss ball appeared in front. I think there were just these two Swiss balls, and I could basically go around moving the block, which was rigid, the cube. And it was hitting these Swiss balls. You could see the Swiss ball deforming a little, jumping around, rolling. All these things were happening.

Claude: Okay, this is actually a really informative result. Let me break down what you observed.

Vivek: Don't you modify prompt one now that I gave you the that I told you what was happening, you can go and modify prompt one.

Claude: Done. Updated Prompt 1 with three changes based on what you saw.

Genie Iteration: The Silver Medical Probe

Vivek: Okay. So I think it's still it's still, you know, creating the sketch and then we'll see what app oh, wait. Look. Something happened. What did you say? Let me look at the prompts again. A small handheld block shaped sensor roughly the size of a smartphone with a polished spherical tape protruding from one end. The sensor floats at human okay. It created some oh, that looks disgusting. It it basically created some disgusting looking big silver box

with a tiny metal probe. It it looks very disgusting.

Claude: Yeah, that's my fault — the phrase "polished spherical tip protruding from one end" cued Genie to render a medical probe aesthetic, which looks clinical and gross. Let me fix that.

Genie Iteration: Scale Failure

Vivek: Okay. Now it's making a very small hemisphere. And a ridiculously large iPhone, like, sensor which is bigger than the and probing it. That doesn't make any science sense either.

Claude: Yeah, that's on me again. My previous prompt said "hemisphere 10 cm across" and "sensor size of an iPhone" — an iPhone is bigger than 10 cm, so I literally told Genie the sensor is bigger than the hemisphere. It did exactly what I asked. My bad.

Vivek: No. Now it looks like a very weird now it look you remember those phones from the nineties? Like, those ones you would pick up and start talking? It looks like partly that, partly, like, some medical device that with a probe that's, like, stuck to the surface of the sphere.

Claude: Okay, I figured out what's happening. Every time I describe the sensor with a "tip" or "probe" protruding from the end, Genie's training data pulls from medical-device imagery — because that's the only thing in its training data that looks like a "handheld thing with a protruding tip pressed against soft tissue." And the lab environment reinforces it.

Genie Iteration: The 90s Cordless Phone

Vivek: Oh, wow. That's now what it's done is it has a there's a hemisphere, the hemisphere is deforming without a sensor contacting it. Like, a weird UFO, like, Apple product hovering over the surface. Like, you know, like, a weird Apple it looks like a app new Apple product, which is a maybe a earphone's case, which is like a hovering over the, uh, surface. And there's a dent in the hemisphere. Where it's hovering over it. It's not even in contact.

Claude: Okay, this is actually really telling about how Genie works. Quick diagnosis question first: is this the static PREVIEW frame — the one it shows before you click "explore" — or are you already in the interactive session moving the sensor around with WASD?

Vivek: Yeah. So this is preview. Like, I haven't generated the world yet. Should I just are you just saying should I just click create world and then figure out go and interact with the world and then tell you what's happening.

Claude: Yes, exactly. Just click create world. Preview frames are notorious for being weird — they're one-shot generated stills, not representative of interactive behavior. Once you're in-world and using WASD or mouse to move the puck around, Genie will

generate new frames in response to your inputs, and contact should look much more sensible. Go explore, hit a few points on the hemisphere, and come back with what you see. Don't judge the prompt based on the preview.

Vivek: Okay. So you know those, like, those I don't know if they're, like, ice hockey or you go to those shopping malls and those showrooms, and you have those games where two people hit a puck around. You know, like, with the it's like a board where the puck moves very fast and like a cylindrical disc that jumps around and just moves around in arena. Like a small table like arena. So it looked like you had a hemisphere that was mounted on a table like arena, and then basically, instead of a human playing a game where he hits the puck, I am it felt like I was playing a video game instead of me being in that shopping mall hitting that I'm basically controlling this from keyboard, and I can make it jump up and down use WASD to have the puck move around. And the it doesn't, like, slide on the surface or apply any pressure. It's like a it's like a game where you literally go to these malls and you hit the puck. It's Except that I was not in a designer showroom or a or a Apple shop in the mall. I was basically looking at a Apple shop in a mall from my computer, and I was or, like, uh, you know, these malls have those showrooms where you have those games with the and you hit them. And I was basically hitting the puck and having it jump around using my keyboard. That's what happened.

Claude: Okay, big diagnosis from your last description: the word "puck" was a disaster. Genie latched onto air-hockey imagery because "puck" plus "arena" in its training data strongly cues sports games. And "display plinth" got interpreted as a game table. That's why you got a video-game-in-a-mall experience.

Genie Iteration: The Chicken iPhone

Vivek: Oh, this is this went horribly wrong. Now it basically drew a hemisphere like, you know, like chicken, like, you know, when you want to cut chicken in the kitchen, you you go wash it and you're late. It looked like a hemisphere of chicken meat. And there was a human hand trying to cut the hemispherical chicken meat with an iPhone that I couldn't even control. Like, even the WASD wasn't working. It was literally a human hand using an iPhone to cut a hemisphere which looked like chicken meat. I even know what to say any longer.

Claude: Oh my god. Okay. That is genuinely hilarious. A hand cutting a chicken hemisphere with an iPhone. I think I literally just prompted "kitchen countertop" plus "flesh-coloured hemisphere" plus "something being lowered onto it" and Genie's training data said "oh, this is a meat preparation scene, obviously." And the WASD not working means Genie got locked into a pre-scripted animation instead of giving you interactive control. Every single time I try to fix one association, Genie's training data pulls in a completely new weird one.

Vivek: Okay. So here's the thing. Right? Like, the thing is that

Claude: Listening.

Vivek: Yes. I mean, the here's the thing. Right? The thing is that these it looks like I never really used Gemini a lot just because I don't know. I felt like I never enjoyed the experience. I was never good at prompting it. Like, I was very good at prompting ChatGPT and you, Claude. And I can get insane results with when I collaborate with you just building all these I don't know. Was it the other session or this session? I I don't know. Hundred and fifty skills, and we committed everything. I guess it was this session. Right?

Claude: Yeah, the 150 skills was this session. We ran sync-os right before switching to voice. Nice to know you remember the milestone — 149 to 150 happened in the same sitting as building find-papers-by-similar-authors.

Vivek: Okay. So let let's do one thing. So one thing to take note of is that I think we're at a stage where, you know, I wanted to sort of do a wise discussion of these two papers mainly and going you know, what's gonna happen going forward or my thoughts about those two papers that I thought the exploration one and the Chen tactile tomography one, which I thought were about local you know, building global stiffness maps by stitching together local estimates, which we aren't really interested in. They came out both in twenty twenty six, which explains why I didn't hit them in my previous literature review. It validates my previous literature review, And the real main thing is about figuring out if they have good instrumentation. One paper just led to another paper that we need to investigate. And then the other paper might be worth looking into, and that's where you have all the that that F/T sensor. Right? And that led to this whole discussion on you know, what's the best sensor, do we go about instrumentation, and that led to the Genie prototyping thing. I'll say that just to wrap up the voice session part, or we'll continue talking, but just to wrap up, what I had initially thought of was that it was a discussion about these two papers that were from local to global. They came out in twenty twenty six, and basically, what is of interest to us from those papers is the instrumentation, and we're trying to think about the best way to get a visceral sense for what's the best instrumentation, how do we prototype it, how do we visualize it, It's it's we really now brainstorming about the instrumentation, and that's the sort of next steps from these papers.

Claude: Yeah, that's a really clean wrap-up and I think your synthesis is exactly right. Let me mirror it back cleanly.